12.6 Solving Multi-Step Problems Involving Discounts, Taxes, Tips, or Fees

Lesson Introduction

Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.	Learning Target	• I can solve multi-step problems that involve discounts, markups, taxes, tips, or fees.

	Building On	Working Toward
Benchmark Coherence	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.	N/A

Essential Questions	• How do you solve multi-step problems that involve discounts, markups, taxes, tips, or fees? • How can you determine the amount of sales tax or percent gratuity given a bill subtotal and final total?
Lesson Narrative	In this lesson, students solve multi-step real-world problems involving discounts, markups, taxes, tips, or fees. This lesson builds on students' prior work solving one-step discount, markup, tax, tip, and fee problems.

Component Summary	12.6.1 Warm-Up (~5 min.)	Consider percentages on a double number-line model *(SE p. 241)*	12.6.3 Collaboration (10-12 min.)	Collaboratively solve problems involving taxes, tips, and totals *(SE p. 243)*
	12.6.2 Exploration (10-15 min.)	Explore finding sales tax rates and tipping in real-world contexts *(SE p. 242)*	12.6.4 Your Turn! (10-13 min.)	Practice solving problems involving taxes, tips, fees, discounts, and markups *(SE p. 245)*
	12.6.5 Wrap-Up (~5 min.)	Demonstrate understanding of solving multi-step sales tax, discount, markup, and fee problems *(SE p. 246)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: N/A <u>Additional Terms</u>: • Gratuity (Tip) • Markup • Discount • Sales Tax • Subtotal • Total	• Calculators • (Optional) Chart paper • (Optional) Markers	• Consider which problems or components you may want students to solve without using a calculator. • Consider what calculations/work you want students to show in their workbook for problems where they use a calculator.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle converting between different forms of rational numbers (i.e., fraction to decimal, percent to decimal, decimal to percent).
Students may not understand the difference between the subtotal and the total and which are both used when determining tax or tip amounts in various contexts.</td><td>During the 12.6.2 Exploration, consider modeling a portion of problem if needed, providing specific instructions on strategies that can be used to find a sales tax percentage. Listen as students collaborate on question 1a to determine if this is necessary.
During the 12.6.4 Your Turn!, facilitate a discussion with students explaining that tips can be based on the subtotal (without tax) or based off the total (with taxes).
Allowing students to use calculators during the lesson will help students focus on discovery and process of calculating discounts, markups, taxes, tips, and percentages. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers.
There are multiple strategies for solving the problems in this lesson. Encourage students to use strategies that make sense for them as they build fluency in solving problems.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think Pair Share
In 12.6.3 Collaboration, consider asking students to initially complete the problems independently. Then have students discuss their work with a partner. These discussions will help students gain confidence and understanding.</td><td>MA.K12.MTR.6.1: Reasonableness
In 12.6.4 Your Turn!, students solve multi-step problems using a variety of strategies. Prompt students to informally check their answers by considering if the solution makes sense given the context. Then have students also formally check their calculations for accuracy.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider building upon any anchor charts created in earlier lessons to add additional content learned.

Consider modifying the 12.6.3 Collaboration and 12.6.4 Your Turn! into stations that students can work through with a partner or small group. Determine how to set up the rotation and which stations may require more support from the teacher as students work through them.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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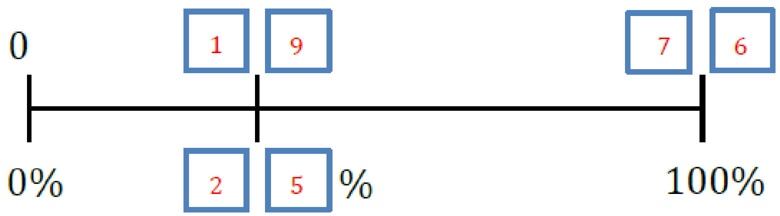
12.6.1 Warm-Up: Percentages on a Linear Model

Component Overview: *Students create an accurate number line using percentages of a whole by filling the digits one through nine into the boxes marked on the number line.*

12.6.1 Warm-Up: Percentages on a Linear Model

1. Fill in the blue boxes using the digits $1-9$, one time each at most, to create an accurate number line.



Students have worked with double number-line models before.

If time allows, consider having students discuss question 2b and/or share strategies/solutions for question 1.

2. There are multiple possible solutions.

   a. Find at least one other solution.
      $17, 68, 25\%$

   b. Explain whether using 47% for the percentage makes sense.
      No, because the line would need to be closer to the middle since 47 is close to 50 which is halfway.

12.6.2 Exploration: Finding Sales Tax Rates and Total Amounts

Component Overview: *Students explore how to find sales tax rates and how to tip in a real-world context.*

12.6.2 Exploration: Finding Sales Tax Rates and Total Amounts

1. Cities often have different sales tax rates based on local government decisions.

   a. Discuss with your partner how you might find the percentage of sales tax if you are given the price of an item and the amount of sales tax paid for the item. Summarize your discussion.
   If you divide the sales tax by the price, you can find the rate. You could also set up a percent proportion to find the rate.

   b. With your partner, complete the two tables to determine the sales tax rates for City A and City B.

City A			
Item	**Price ($)**	**Sales Tax ($)**	**Sales Tax Rate (%)**
Paper Towels	8.99	0.63	7%
Laundry Soap	11.83	0.83	7%

City B			
Item	**Price ($)**	**Sales Tax ($)**	**Sales Tax Rate (%)**
Paper Towels	8.99	0.54	6%
Laundry Soap	11.83	0.71	6%

   c. Determine the total cost, with tax, of paper towels and laundry soap in City A.
   $22.28

   d. The pre-tax cost of 5 beach towels in City A is $54.95. Determine the total cost, with tax, of 5 beach towels in City A.
   $58.80

If needed, consider doing a Think Aloud as you solve question 1b (only City A) after giving students an opportunity to discuss question 1a. This will provide a model for students as they try the other problems.

Is the sales tax rate for both items in one city the same or different? Why would this make sense in context?

12.6.2 Exploration: Finding Sales Tax Rates and Total Amounts (continued)

e. If the subtotal for the purchase of a new bathing suit in City B is $47.97, what is the total cost of the bathing suit with tax?
$50.85

f. Find the total cost, with tax, on paper towels, laundry soap, and a new bathing suit in City B.
$72.92

2. Jada and Corrin had lunch at a local restaurant. The final bill for their meal is shown.

They decide to tip their server 18% of the subtotal. Jada says the tip should be $7.56, but Corrin disagrees. Corrin calculated the tip to be $8.28.

Date: Sep. 12th
Time: 6:55 PM
Server: # 27

Bread Stix	9.50
Chicken Parm	15.50
Chef Salad	12.00
Lemon Soda	2.00
Tea	3.00
Subtotal	42.00
Sales Tax	3.99
Total	45.99

a. Explain who calculated the correct tip.
Jada correctly calculated the tip because $42.00 times 0.18 is $7.56. Corrin calculated the tip from the total which includes taxes, instead of from the subtotal.

b. What is the total amount, including the tip, that the friends paid at the restaurant?
$53.55

Prompt students to calculate the tip and then compare their answers to Jada's and Corrin's answers. Remind students to convert the tip percentage to a decimal.

12.6.3 Collaboration: Taxes, Tips, and Totals

Component Overview: *Students collaboratively determine taxes, tips, markups, and discounts in real-world contexts.*

12.6.3 Collaboration: Taxes, Tips, and Totals

Work through each situation with your partner. Be sure to show your work.

1. Xavier visits an Italian restaurant for dinner.

   a. If Xavier tips 20% on the subtotal $24.95, how much tip will he pay?
   $4.99

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Date: Sep 19th
Time: 6:04 PM
Server: Flo

Bread Stix        9.50
Ravioli Bites    10.50
Cheesecake        4.95

Subtotal         24.95
Sales Tax        ____
Tip              ____
Total            ____
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   b. The city has a sales tax rate of 9.5%. How much in tax will the customer pay on the subtotal of $24.95?
   $2.37

   c. What is the total Xavier spent on his meal, including tip and tax?
   $32.31

Think-Pair-Share
Ask students to initially complete the questions independently. Then have students discuss their work with a partner. These discussions will help students gain confidence and understanding.

Consider asking students the following:

- *Does the sales tax amount depend on the tip amount? Or are those two separate, additional costs?*
- *Are these amounts related to the subtotal or the total cost?*

12.6.3 Collaboration: Taxes, Tips, and Totals (continued)

2. A sporting goods store purchased baseball hats from a supplier for $\$24$ each and marked them up to $\$35$.

   a. Calculate the percentage of the markup.
   46%

   b. A customer orders 4 baseball hats from the store online. What is the subtotal of the 4 hats?
   $140.00

   c. If the sales tax is 6.25% of the subtotal and there is a $\$7.00$ shipping fee applied after tax, what is the total amount the customer paid?
   $155.75

3. Leilani found a 30% off discount code for an NXT True Hybrid Surfboard. The original cost of the surfboard is $\$700$. If the company does not charge tax but instead charges a $\$150$ shipping fee, how much does the surfboard cost in total?
   $640.00

Consider having students write their work on individual dry-erase boards to make sharing their work during the debrief easier.

12.6.4 Your Turn!

Component Overview: *Students practice completing multi-step problems involving taxes, tips, fees, markups, and discounts.*

12.6.4 Your Turn!

1. A car dealership pays a wholesale price of $12,000 to purchase a vehicle. The car dealership wants to make a 32% profit.

   a. How much money is added to the wholesale price to make a 32% profit?
   $3,840

   b. After the markup, what is the total retail price of the vehicle?
   $15,840

   c. The sales tax in the city it is sold is 7.5%. After tax, what will the customer pay for the car?
   $17,028

2. The table shows the pre-tax cost of items at a convenience store where the sales tax rate is 4.6%.

Item	Cost ($)
Milk	3.75
Bag of Ice	2.50
Aspirin – 24 tablets	8.25
Snack Mix	3.75
Candy Bar	1.25
Batteries	6.75

   a. Cal stopped at the convenience store and picked up a bag of ice and batteries. What was his total cost, with tax?
   $9.68

Consider combining the 12.6.3 Collaboration and Your Turn! into stations and having students rotate through the problems.

12.6.4 Your Turn! (continued)

b. Ms. Isales has $10.00 in her wallet. Does she have enough to buy aspirin and a candy bar? Explain your reasoning using mathematics.
Yes, because when you add the items and find the tax it adds up to $9.94 which is less than $10.

3. A local restaurant automatically adds a 21% gratuity (tip), pre-tax, to each bill for parties over 10. The table shows the bills due for several parties over 10. Complete the table.

Party	Subtotal ($)	Sales Tax ($)	Gratuity ($)	Total Amount of Bill
Flores Party	93.75	5.86	$19.69	$119.30
Atkinson Party	176.99	$11.06	37.17	$225.22
Mikelson Party	135.50	$8.47	$28.46	172.43

Consider asking students the following:
- *What is gratuity? What does it mean that gratuity is pretax?*
- *How can you find the sales tax percentage from the sales tax and subtotal?*
- *Can you find the sales tax and gratuity separately?*

12.6.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students solve a multi-step problem involving a discount, sales tax, and shipping fee. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should be allowed to access a calculator to complete the Wrap-Up.*

12.6.5 Wrap-Up: Ticket Out the Door

1. Diesel has been waiting patiently for the Evolve Carbon All Terrain Skateboard to go on sale. For his birthday, he received a 25% off discount. The original price of the skateboard is $1,999.99 and the company has a 5% sales tax, along with a flat fee of $35 for shipping.

   What is the total amount Diesel needs to purchase his dream skateboard?
   $1,609.99

Challenge students who finish early to write their own multi-step problem. Then have students exchange problems with a partner and solve each other's problem.